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Roll No

AU-702(B)-CBGS

B.Tech., VII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Heat and Mass Transfer

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Write down the significance of Reynold's and Nusselt number? 7
b) What is a radiation shield? 7
2. a) What is intensity of radiation? 7
b) What are the use of heat pipe? 7
3. a) Counter flow heat exchanger most preferred why? 7
b) Give the classification of heat exchangers. 7
4. a) Define effectiveness of a heat exchanger. Derive an expression for the effectiveness of a double pipe parallel flow heat exchanger. State the assumptions made. 7
b) A parallel flow heat exchanger has hot and cold water stream running through it, the flow rates are 10 and 25 kg/min respectively. Inlet temperatures are 75° C and 25° C on hot and cold sides. The exit temperature on the hot side should not exceed 50° C. Assume $h_i = h_o = 600\text{W/m}^2\text{K}$. Calculate the area of heat exchanger using E-NTU approach. 7

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- 5. a) Define emissivity, absorptivity and reflectivity. 7
b) Describe the phenomenon of radiation from real surfaces. 7
- 6. a) State and prove the Kirchhoff's law of radiation. 7
b) State and prove the Stefan - Boltzmann law. 7
- 7. List the dimensionless numbers. Explain any two in detail. 14
- 8. a) What is Fourier's Law of heat conduction? Explain with suitable example 7
b) What is Newtonian heating or cooling process? 7
